Supplementary Material for Texture Resonance Retrieval for Retrieval-Grounded Editable Audio Effect Preset Selection

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S1 Canonical Evidence Sources and Metric Notes

This supplementary package is synchronized to the repository's canonical experimental artifacts and the P0 experiment record. The primary objective evidence consists of the Protocol-A split audit, the earlier resolved-audio-grouped Protocol-A stats, and the P0 E2/E3 summaries that share a 204-query test split. We include the canonical listening-study report as complementary subjective evidence. We do not use the legacy comparison report or the deprecated Protocol-B outputs as formal evidence in this revision.

All objective metrics follow the same operational definitions used in the main paper. **Param. Dist.** is RMSE over flattened numeric parameter leaves; **Acc@0.1** measures the fraction of numeric leaves within an absolute tolerance of 0.1; **Recall** measures retrieval over active numeric parameters; **Cosine** is cosine similarity in the flattened parameter space; and **Module** is a schema-level Jaccard index over active DSP modules inferred from keys ending with `On`. Missing numeric keys are treated as zeros, non-numeric leaves are ignored, and we do not apply per-parameter normalization. Accordingly, absolute magnitudes should be interpreted *within* each stated protocol rather than across protocols.

S2 Protocol-A: Split Audit and Benchmark Definition

The external benchmark drop-in contains 1267 total records. A post-hoc audit of the original song-name-based held-out list (used in earlier drafts) revealed exact cross-split audio reuse, so the manuscript now anchors its main objective evidence to a stricter resolved-audio-grouped split. Table S1 summarizes both definitions.

Table S1: Protocol-A split audit summary. "Near-duplicate" counts come from the repository's cheap fingerprint audit and therefore indicate a remaining risk surface rather than a formal perceptual-duplicate guarantee.

Split definition	$N_{\mathbf{test}}$	$N_{\mathbf{kb}}$	**Shared audio paths**	**Near-duplicate pairs**
Legacy song-name split	211	1056	16	48
Resolved-audio grouped split	204	1063	0	200

The stricter split groups records by resolved local audio path before sampling the held-out set, thereby eliminating exact cross-split audio reuse under the current audit. At the same time, the benchmark remains a pilot setting: the cheap fingerprint still reports many near-duplicate pairs, and the corpus remains guitar-only. The dataset JSON exposes `SongName`, `Style`, `Feature`, `Parameters`, and `Vectors`, but it does not include a separate auditable text-source field; accordingly, `Style` and `Feature` are the only explicit text-side provenance fields available in-repo. The stricter split contains 204 paired TRR rows but only 203 unique `query_name` labels because `Hard Rock Crunch` appears twice in the dataset; all pairing therefore uses `query_idx`.

S3 P0 Protocol-A: Direct Retrieval Comparison

P0 Protocol-A compares TRR with Wav2Vec, FeatureNN, CLAP, PaSST, and PANNs on a shared 204-query split and a 1,063-record retrieval database. All methods use top-1 retrieval. This is the main baseline matrix used by the revised manuscript. The earlier resolved-audio-grouped Protocol-A table is retained below as split-audit context, not as the only main result table.

Table S2: P0 Protocol-A direct retrieval comparison. Lower is better for L2 and Norm.L2; higher is better for the other metrics.

Method	n	L2 ↓	Norm.L2 ↓	Acc@0.1 ↑	Recall ↑	Cosine ↑	Module ↑
TRR (Ours)	204	**8.7287**	**0.1881**	**0.5766**	**0.5404**	**0.8415**	**0.8297**
Wav2Vec	204	22.5174	0.2400	0.5015	0.4648	0.5642	0.7423
FeatureNN	204	17.9391	0.2236	0.5264	0.4779	0.6030	0.7544
CLAP	204	16.1257	0.2298	0.5389	0.4922	0.6531	0.7837
PaSST	204	18.2860	0.2132	0.5422	0.5001	0.6192	0.7960
PANNs	204	21.0269	0.2260	0.5090	0.4759	0.5826	0.7606

S4 P0 Protocol-A: Statistical Reporting

The P0 statistical report tests TRR against each baseline on per-query Norm.L2 using Wilcoxon signed-rank tests with Holm correction. The tests consistently reject the null hypothesis, but the effect sizes are small; the correct interpretation is statistically reliable but moderate TRR advantage.

Table S3: P0 Protocol-A: TRR versus baseline statistical tests on per-query Norm.L2.

Pair	n	p-value	Test	Cohen's d	Holm reject
TRR vs Wav2Vec	204	1.564×10^{-7}	Wilcoxon	-0.2101	Yes
TRR vs FeatureNN	204	9.523×10^{-5}	Wilcoxon	-0.1380	Yes
TRR vs CLAP	204	4.476×10^{-4}	Wilcoxon	-0.2189	Yes
TRR vs PaSST	204	1.254×10^{-3}	Wilcoxon	-0.0980	Yes
TRR vs PANNs	204	2.122×10^{-5}	Wilcoxon	-0.1528	Yes

S5 TRR Mechanism Ablations

The main P0 table includes one mechanism-relevant control: TRR and Wav2Vec use the same Wav2Vec2 acoustic backbone, but differ in the aggregation operator. Wav2Vec uses a first-order mean-pooled embedding, whereas TRR uses projected frame activations followed by Gram-matrix second-order statistics and L2 normalization. Table S4 isolates this comparison from the broader external-baseline matrix.

Table S4: Backbone-controlled diagnostic from P0 Protocol-A. TRR and Wav2Vec share the Wav2Vec2 backbone but use different aggregation operators.

Method	Aggregation	Norm.L2 ↓	Acc@0.1 ↑	Recall ↑	Cosine ↑	Module ↑
TRR	Gram second-order statistics	0.1881	0.5766	0.5404	0.8415	0.8297
Wav2Vec	Mean pooling	0.2400	0.5015	0.4648	0.5642	0.7423

To further test the mechanism rather than only the final leaderboard, we recomputed controlled Wav2Vec2 layer features on the same 204-query P0 split with a fixed random seed and compared same-backbone aggregation variants. This ablation is not a replacement for the user-verified P0 E2 leaderboard because it uses the current local execution path and deterministic projection matrices. Its role is narrower: it checks whether the second-order design choices are directionally supported.

Table S5: Controlled TRR mechanism ablation on the P0 split using the current local execution path. Lower is better for Norm.L2; higher is better for all other metrics.

Variant	**Layers**	**Dim**	**Norm.L2 ↓**	**Acc@0.1 ↑**	**Recall ↑**	**Cosine ↑**
Mean pooling	5	768	0.2582	0.6089	0.4623	0.5895
Gram, no projection	5	589824	0.2501	0.6103	0.4635	0.6092
Gram, projected	4	4096	0.2528	0.6017	0.4585	0.6191
Gram, projected	5	4096	**0.2457**	**0.6104**	**0.4654**	0.6224
Gram, projected	6	4096	0.2529	0.5985	0.4485	0.6317
Gram, projected	4+5+6, $d = 16$	256	0.2501	0.6052	0.4449	0.6070
Gram, projected	4+5+6, $d = 32$	1024	0.2480	0.6022	0.4640	**0.6398**
Gram, projected	4+5+6, $d = 64$	4096	0.2546	0.5963	0.4487	0.6249
Gram, projected	4+5+6, $d = 128$	16384	0.2576	0.6054	0.4538	0.6121
Gram, no L2 norm	4+5+6, $d = 64$	4096	0.2849	0.5645	0.3944	0.5857

The mechanism result supports three bounded claims. First, Gram aggregation is not merely a Wav2Vec2-backbone effect: the same-layer Gram variants improve normalized distance over same-backbone mean pooling. Second, normalization is important; removing L2 normalization produces the weakest row in the table. Third, architectural choices remain meaningful: layer 5 and $d = 32/d = 64$ projected multi-layer variants trade off different metrics, so the paper should not claim that the reported TRR configuration is globally optimal. The generated artifact also records top-3 and top-5 oracle-best normalized L2 within each retrieval list, which we use only as a top-K sensitivity diagnostic rather than as the deployed top-1 metric. We therefore frame TRR as a useful texture-aware retrieval prior rather than as a fully optimized representation family.

S6 Resolved-Audio-Grouped Protocol-A Audit Context

Before the P0 baseline matrix was added, the revision used a resolved-audio-grouped Protocol-A table to avoid exact cross-split audio reuse. That table remains useful as audit context because it documents the anti-leakage split behavior and the earlier CLAP-overlap comparison. It should not be mixed numerically with the P0 table because the reported L2 scale and baseline set differ.

Table S6: Resolved-audio-grouped Protocol-A audit table retained for split transparency.

Method	n	**Param. Dist.**	**Acc@0.1**	**Recall**	**Cosine**	**Module**
TRR (Ours)	204	8.0467	0.5169	0.4595	0.8247	0.8051
Text-RAG	204	24.1705	0.4717	0.3967	0.4943	0.7456
Wav2Vec-RAG	204	23.8197	0.4251	0.3505	0.5292	0.7043
FeatureNN-RAG	204	19.2613	0.4198	0.3600	0.5479	0.7134
CLAP	201	22.3102	0.4492	0.3971	0.5725	0.7240

S7 Protocol-A: Per-Query Diagnostics

To make the main benchmark less dependent on mean-only reporting, we derive per-query diagnostics from the canonical Protocol-A CSV for the resolved-audio-grouped split. The source artifact is the derived Protocol-A analysis file pair in Markdown and JSON form.

These diagnostics are paired by `query_idx`. The TRR rows contain 204 paired queries but only 203 unique `query_name` labels because one label (`Hard Rock Crunch`) is reused in the CSV; we therefore treat `query_idx` as the authoritative pairing key.

Table S7: Per-query win/loss/tie summary for TRR under Protocol-A. The interquartile range collapses to zero in many rows because several queries retrieve the same preset under both methods.

Baseline	Metric	Wins	Losses	Ties	Mean Δ
CLAP	Param. Dist.	130	54	17	14.1460
CLAP	Cosine	124	60	17	0.2503
CLAP	Module	79	39	83	0.0781
FeatureNN-RAG	Param. Dist.	130	49	25	11.2146
FeatureNN-RAG	Cosine	122	57	25	0.2769
FeatureNN-RAG	Module	86	40	78	0.0917
Text-RAG	Param. Dist.	117	69	18	16.1238
Text-RAG	Cosine	117	69	18	0.3304
Text-RAG	Module	73	56	75	0.0595
Wav2Vec-RAG	Param. Dist.	125	40	39	15.7730
Wav2Vec-RAG	Cosine	121	44	39	0.2956
Wav2Vec-RAG	Module	84	43	77	0.1008

Table S8: Representative Protocol-A L2 cases from the derived per-query analysis. Positive Δ indicates TRR is better.

Type	Baseline	Query	TRR retrieved	Baseline retrieved	ΔL2
Largest gain	CLAP	Modular Hybrid Drift	Prepared Guitar Static	Ethereal Wave	+71.6821
Largest gain	FeatureNN-RAG	Sub-Bass Drone	Hauntology Library	Dark Ambient Drone Prism	+72.1369
Largest gain	Text-RAG	Sub-Bass Drone	Hauntology Library	Studio Clean	+72.2040
Largest gain	Wav2Vec-RAG	3-Chord Punk	Distorted Cassette	Britpop Jangle	+76.2671
Largest failure	CLAP	Talk Box Funk	Schlager Pop	Drum and Bass Synth	-57.3198
Largest failure	FeatureNN-RAG	Rave Hardcore	Dark Ambient	Melodic Death Metal	-56.3859
Largest failure	Text-RAG	Talk Box Funk	Schlager Pop	Studio Clean	-57.3623
Largest failure	Wav2Vec-RAG	Phased Synth	London Jazz	Nightcore Speed	-51.6878

S8 P0 Protocol-C: Real-Degradation Robustness

P0 Protocol-C evaluates realistic audio degradations by corrupting raw audio first and then re-encoding the degraded signal into TRR embeddings. It reuses the P0 Protocol-A 204-query test split, uses `Style+Feature` as the text branch, and computes ranking relevance from parameter-space oracle neighbors rather than exact song-name matching.

Table S9: P0 Protocol-C overall summary on the shared 204-query split. Lower is better for L2; higher is better for all other metrics.

Method	α	L2 $\downarrow$	Acc@0.1 $\uparrow$	MRR $\uparrow$	NDCG@5 $\uparrow$
fixed fusion 0.30	0.3000	29.2322	0.4846	0.4095	0.3723
fixed fusion 0.50	0.5000	20.5551	0.6085	0.5607	0.5528
fixed fusion 0.70	0.7000	20.1868	0.6104	0.5652	0.5551
adaptive fusion	0.5980	**20.1491**	**0.6121**	**0.5654**	**0.5566**

Table S10: P0 Protocol-C per-condition winners across the main multimodal comparison set.

Condition	Best L2	Best Acc@0.1	Best MRR	Best NDCG@5
AWGN 20 dB	fixed 0.70	adaptive	adaptive	adaptive
AWGN 10 dB	fixed 0.70	adaptive	adaptive / fixed 0.50	fixed 0.50
AWGN 5 dB	fixed 0.70	fixed 0.70	fixed 0.70	fixed 0.70
MP3 128k	fixed 0.70	fixed 0.50	fixed 0.50	fixed 0.50
MP3 64k	fixed 0.70	fixed 0.50	fixed 0.50	fixed 0.50
MP3 32k	adaptive	fixed 0.50	adaptive / fixed 0.50	fixed 0.50
Reverb 0.6	adaptive	fixed 0.50	adaptive / fixed 0.50	adaptive
Reverb 1.0	adaptive	fixed 0.50	adaptive / fixed 0.50	fixed 0.50
Truncate 50%	adaptive	adaptive	adaptive / fixed 0.70	fixed 0.70
Truncate 30%	adaptive / fixed 0.70	adaptive	adaptive / fixed 0.70	adaptive / fixed 0.70

The per-condition view shows that adaptive fusion is not a trivial uniform winner; fixed 0.50 and fixed 0.70 remain competitive on several conditions. The stronger conclusion is that audio-heavy fusion is fragile, whereas adaptive fusion consistently moves the system toward the stable mid-to-high text-weight region while preserving multimodal behavior.

Local boundary audit. We additionally ran the current local Protocol-C script with explicit audio-only ($\alpha = 0.0$), pure-text ($\alpha = 1.0$), and parameter-oracle upper-bound rows. Because this run uses the current repository execution path rather than the server-side P0 artifact snapshot, we report it as an audit boundary rather than as a replacement for Table S9. The result in Table S11 confirms the main qualitative interpretation: audio-only retrieval is weak under realistic degradation, mid/high text weights are stable, and the pure-text boundary remains a strong practical boundary. The oracle row is not a method; it selects the nearest KB item using ground-truth parameters and only quantifies remaining headroom. Thus, adaptive fusion should be read as a multimodal operating strategy near the stable region, not as proof of superiority over text-only retrieval.

Table S11: Current-code Protocol-C boundary audit with audio-only and pure-text rows. This table is an execution-path audit artifact, not a replacement for the canonical P0 server summary.

Method	α	L2 $\downarrow$	Acc@0.1 $\uparrow$	MRR $\uparrow$	NDCG@5 $\uparrow$
audio-only	0.0000	102.5577	0.4962	0.1184	0.0731
fixed fusion 0.30	0.3000	81.0390	0.5980	0.3916	0.3588
fixed fusion 0.50	0.5000	66.9431	0.6791	0.5633	0.5463
fixed fusion 0.70	0.7000	63.9030	0.6832	0.5730	0.5528
adaptive fusion	0.5980	65.4614	0.6837	0.5719	0.5529
pure text	1.0000	**61.5079**	**0.6872**	**0.5847**	**0.5586**
parameter oracle upper bound	oracle	1.3284	0.9548	1.0000	1.0000

S9 Listening Study Statistics

The listening study uses a multiple-stimulus protocol with a hidden reference and no explicit low-quality anchor. The raw logs label our system as `HCAP`; throughout the manuscript and this supplement we refer to the same condition as the *TRR-based system*. Across all trials, the study contains 26 unique participants and 910 ratings.

The current logs preserve repeated-measures structure and participant IDs, but they do not preserve device metadata, participant expertise labels, or the provenance and time budget of the manual baseline. We therefore interpret Trials 2–5 as perceptual context rather than as a fairness-critical human-vs-system optimization benchmark.

Table S12: Study-level listening-test summary.

Trial block	Rows	Participants	Stimuli	Raw mean	Raw median / std
Trial 1	208	26	8	72.92	77.50 / 22.72
Trials 2–5	312	26	3	69.53	73.50 / 23.53
Trials 6–10	390	26	3	57.54	59.00 / 32.24

Table S13: Repeated-measures nonparametric tests for the listening study.

Block	Comparison	n	Mean(A-B)	Median(A-B)	rbc	p (Holm)
Trials 2–5	Friedman omnibus ($Q = 41.6154$, $p_{\mathrm{perm}} = 5 \times 10^{-5}$)	N/A	N/A	N/A	N/A	N/A
Trials 2–5	reference vs TRR-based system	26	13.78	8.38	0.860	0.00015
Trials 2–5	reference vs manual	26	33.61	26.00	0.983	0.00015
Trials 2–5	TRR-based system vs manual	26	19.83	17.25	1.000	0.00015
Trials 6–10	Friedman omnibus ($Q = 36.2308$, $p_{\mathrm{perm}} = 5 \times 10^{-5}$)	N/A	N/A	N/A	N/A	N/A
Trials 6–10	reference vs MusicGen	26	47.72	50.90	1.000	0.00015
Trials 6–10	reference vs TRR-based system	26	46.71	51.30	0.994	0.00015
Trials 6–10	MusicGen vs TRR-based system	24	-1.10	0.80	0.067	0.7797

Table S14: Trial 1 style-matching stimulus labels used in the listening study.

Label	Interpretation
Jazz Clean	clean jazz tone target
Blues Solo	blues lead tone target
Phase	phaser/phase-effect target
Ambient Guitar	ambient guitar texture target
Flanger	flanger-effect target
Modern Metal	modern high-gain tone target
Chorus	chorus-effect target

S10 Near-Duplicate Sensitivity Analysis

To assess the impact of near-duplicate presets on retrieval metrics, we progressively remove near-duplicates from the knowledge base using normalized parameter-space RMSE thresholds and re-evaluate locally available stored-vector retrieval methods under the P0 deterministic split. This is a sensitivity analysis rather than the main P0 leaderboard, because the local JSON used for this run contains CLAP vectors for 166 of the 204 test queries and does not contain local PaSST/PANNs vectors. Table S15 reports the resulting robustness pattern.

Table S15: Near-duplicate sensitivity analysis under normalized parameter-space RMSE deduplication. KB size decreases as the threshold τ increases. Param. Dist. $\downarrow$; all other metrics $\uparrow$.

Method	τ	**KB size**	**Param. Dist.** $\downarrow$	**Acc@0.1** $\uparrow$	**Recall** $\uparrow$	**Cosine** $\uparrow$	**Module** $\uparrow$
$\tau = 0$ *(full KB, baseline)*							
TRR	0	1063	37.9848	0.6605	0.5311	0.8149	0.8212
Wav2Vec	0	1063	74.2198	0.6012	0.4570	0.5895	0.7401
FeatureNN	0	1063	60.3766	0.6320	0.4861	0.6688	0.7582
CLAP	0	1063	69.1914	0.5654	0.3941	0.6276	0.7271
$\tau = 0.005$							
TRR	0.005	469	36.9790	0.6461	0.5103	0.8169	0.7866
Wav2Vec	0.005	469	98.7481	0.5616	0.4254	0.5060	0.7125
FeatureNN	0.005	469	86.6753	0.5838	0.4458	0.5414	0.7127
CLAP	0.005	469	109.5363	0.5036	0.3384	0.4435	0.6619
$\tau = 0.01$							
TRR	0.01	409	35.4043	0.6415	0.5142	0.8442	0.8040
Wav2Vec	0.01	409	103.6916	0.5562	0.4330	0.5113	0.7263
FeatureNN	0.01	409	96.9060	0.5698	0.4421	0.5180	0.7064
CLAP	0.01	409	123.1859	0.4854	0.3385	0.4094	0.6633
$\tau = 0.02$							
TRR	0.02	160	47.7463	0.6203	0.4753	0.7758	0.7612
Wav2Vec	0.02	160	100.7927	0.5352	0.3916	0.5277	0.6986
FeatureNN	0.02	160	102.8913	0.5261	0.3586	0.5082	0.6713
CLAP	0.02	160	125.3818	0.4887	0.3142	0.4414	0.6547
$\tau = 0.05$							
TRR	0.05	83	51.4764	0.6023	0.4405	0.7024	0.7136
Wav2Vec	0.05	83	71.8281	0.5425	0.3663	0.5711	0.6719
FeatureNN	0.05	83	60.3433	0.5470	0.3494	0.5722	0.6472
CLAP	0.05	83	70.0935	0.5346	0.3431	0.6039	0.6840

S11 Additional Baseline Status

P0 closes the PaSST and PANNs retrieval-baseline gap for the shared Protocol-A benchmark. We also add a diagnostic direct-regression baseline using mean-pooled Wav2Vec2 features and a two-layer MLP trained only on the P0 knowledge base. This regressor is useful as a boundary condition, but it is not an executable-preset retriever: it predicts dense continuous parameter leaves and still requires constraint projection before deployment. Learned reranking remains future work.

Table S16: P0 Protocol-A additional baseline status.

Method	n	**Evidence status**	**Use in paper**	**Boundary**
PaSST	204	P0 complete	Main baseline matrix	Small TRR effect size
PANNs	204	P0 complete	Main baseline matrix	Small TRR effect size
MLP-Regressor	204	Diagnostic complete	Boundary baseline	Dense non-executable output

The diagnostic MLP regressor obtains raw L2 62.0612, normalized L2 0.1607, Acc@0.1 0.7884, Recall 0.3209, and Cosine 0.4964 on 204 test queries. This mixed profile is informative: direct regression can reduce normalized numeric error by predicting dense average-like parameter leaves, but the low recall and cosine score indicate that it does not recover the same active parameter structure as retrieval. We therefore use it as a boundary condition and do not merge it into the main retrieval leaderboard.

S12 Trials 2–5: Detailed Per-Trial Results

The main text reports only the aggregate Trials 2–5 results. Table S17 provides per-trial stimulus labels and per-condition ratings for completeness.

Table S17: Trials 2–5 per-trial breakdown. Ratings are raw means $\pm$ standard deviation across 26 participants.

Trial	**Stimulus**	**Reference**	**TRR-based**	**Manual**
Trial 2	Jazz Clean	91.19 ± 10.52	75.12 ± 13.99	53.88 ± 22.90
Trial 3	Blues Solo	85.12 ± 21.50	70.04 ± 17.28	58.54 ± 19.01
Trial 4	Phase	83.58 ± 18.74	66.88 ± 18.94	49.04 ± 19.35
Trial 5	Ambient Guitar	81.42 ± 19.66	74.15 ± 15.57	45.42 ± 25.44